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A Theorem on Reconstruction of Random Graphs

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In this paper we prove that given a finite collection of finite graphs, and the subsets of vertices of a random graph G that induce those graphs, it is almost always possible to uniquely reconstruct a class of graphs equivalent to G.

1. Introduction

Let $\mathscr{F}$ be a finite collection of simple graphs. For any graph $G = (V, E)$, let $H(G)$ be a hypergraph on V whose hyperedges are the subsets of V inducing in G a graph in $\mathscr{F}$. The hypergraph $H = H(G)$ is called the $\mathscr{F}$-structure of G.

Some questions arise if we consider the reverse of this construction, *i.e.*, suppose we are given $H = H(G)$, can we reconstruct G? How many graphs are there having H as $\mathscr{F}$-structure? What is the proportion of graphs that can be reconstructed in a unique way from the information contained in their $\mathscr{F}$-structure?

For example:

1 Let $\mathscr{F} = \{$,  $\}$. Then the $\mathscr{F}$-structure of a graph G is called a *two-graph* by Seidel [8], who showed that it determines G up to the relation of switching defined below.

2 Let $\mathscr{F} = \{$  $\}$. The semi-strong perfect graph theorem (Reed [6]) asserts that the property of perfectness of a graph is determined by its $\mathscr{F}$-structure. As a special case of Theorem 1.1 below, for almost all graphs G, the only other graph having the same $\mathscr{F}$-structure as G is its complement.

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Let G_W denote the subgraph of G induced by $W \subseteq V$. We define five equivalence classes on graphs as follows:

— Equality: two graphs are equal if they have exactly the same edges. Denote by $I(G)$ the equivalence class of G.

— Complementation: the complement of G is the graph $\overline{G}$ whose edges are precisely the non-edges of G. Denote by $D(G)$ the equivalence class of G.

— Switching: G, G' are equivalent under switching if there is a subset of vertices W such that $G_W = G'_W$, $G_{V\setminus W} = G'_{V\setminus W}$ and the $(W, V \setminus W)$-edges (respectively non-edges) of G are non-edges (respectively edges) of G'. Let $S(G)$ be the equivalence class of G under switching.

— Complementation + switching. Let $B(G)$ denote the set of graphs equivalent to G or $\overline{G}$ by switching.

— All. Let $A(G)$ be the set of all graphs with the same vertex set as G.

Automorphisms, anti-automorphisms, switching-automorphisms, switching/anti-automorphisms of G are permutations on V that map G to a graph in I, D, S, B respectively.

The objective of this paper is to show:

Theorem 1.1. *Let $\mathcal{F}$ be a finite collection of finite graphs. Then, for some equivalence class $E \in \{I,D,S,B,A\}$, and almost all finite random graphs G, a graph G' has the same $\mathcal{F}$-structure as G if and only if $G' \in E(G)$.*

2. Random graphs

In a random graph on n vertices, the edges are chosen independently with probability p. If we choose $p = 1/2$, we can identify a random graph with one taken at random from the collection of all graphs on n vertices. Therefore, finding the proportion of graphs with a given property is the same as finding the probability that this property holds for a random graph.

We say that a property Q is true *for almost all graphs* if the probability that a random graph has Q tends to 1 as n tends to infinity. See [1] for some of the many properties of almost all graphs.

To prove Theorem 1.1 we will need some ideas about infinite highly-symmetric graphs. So, consider random graphs on a countable number of vertices. Amongst these, consider those for which, given any two disjoint finite subsets of vertices U and W, there exists a vertex z joined to every vertex in U and to none in W. Erdős and Rényi proved that, surprisingly, there exists only one such graph up to isomorphism. We adopt the Rado's construction [7] as our countable random graph R: the vertex-set is $\mathbb{N}$, vertices x, y being joined if 2^x occurs in the expansion of y in base 2 or vice-versa.

The main properties of R (see [2]) are:

1 R is universal – R contains every finite graph as induced subgraph;

2 R is homogeneous – any isomorphism between finite subgraphs can be extended to an automorphism of R.

The following theorem is a restatement of the result proved by Fagin [4], and in a more graph theoretical approach by Blass and Harary in [1].

Theorem 2.1. *Let Ψ be a first order sentence on the language of graphs. If Ψ holds in R, then Ψ holds in almost all finite graphs.*

The next result, by Thomas [9], gives us more information about $Aut(R)$. A permutation group on a countable set is *closed* if it is the automorphism group of some first-order structure. This is equivalent to being closed in the topology of pointwise convergence (see [3]).

Theorem 2.2. *Let M be a permutation group that contains $Aut(R)$ and is a closed subgroup of $Sym(\mathbb{N})$. Then M is one of the following:*

(i) $Aut(R)$;
(ii) $\mathscr{D}(R)$ – *the group of automorphisms and anti-automorphisms of R;*
(iii) $\mathscr{S}(R)$ – *the group of switching automorphisms of R;*
(iv) $\mathscr{B}(R)$ – *the group of switching automorphisms or anti-automorphisms of R;*
(v) $Sym(\mathbb{N})$.

A permutation group on $\mathbb{N}$ is *oligomorphic* if for each $n \in \mathbb{N}$, there are only finitely many orbits on n-subsets of $\mathbb{N}$. The next result is a special case of Ryll-Nardzewski's theorem (see [3] for a discussion).

Theorem 2.3. *Let $\mathscr{M}$ be a countable first order structure. If $Aut(\mathscr{M})$ is oligomorphic, then for any n-tuples $\mathbf{x}, \mathbf{y}$ the following assertions are equivalent:*

(i) $\mathbf{x}$ *and* $\mathbf{y}$ *lie in the same orbit of $Aut(\mathscr{M})$;*
(ii) $\mathbf{x}$ *and* $\mathbf{y}$ *satisfy the same n-variable first order formulae.*

3. The proof

Let H be the $\mathscr{F}$-structure of R. It is clear that each automorphism of R is also an automorphism of H, so that $Aut(R) \subseteq Aut(H)$. Since $Aut(H)$ is closed, it satisfies the hypothesis of Theorem 2.2.

Now, suppose $\mathscr{F}$ is not invariant under either switching or complementation, *i.e.*, there are graphs $F_1, F_2 \in \mathscr{F}, F_2' \in S(F_2)$ such that $\overline{F}_1, F_2' \notin \mathscr{F}$. If $Aut(H) = \mathscr{D}(R)$, then any anti-automorphism of R maps R to its complement. Since R contains F_1 and $\overline{F}_1$ as induced subgraphs, H contains a hyperedge that induces $\overline{F}_1$, contradicting the assumption. If $Aut(H) = \mathscr{S}(R)$, then there is a switching-automorphism of R mapping F_2 to F_2', and again H contains a hyperedge inducing F_2' in R. Hence $Aut(H) = Aut(R)$. Similarly, if $\mathscr{F}$ is invariant under complementation but not switching, then $Aut(H) = \mathscr{D}(R)$, and so on. Therefore there are only five possibilities for $Aut(H)$, and the right one is determined by the collection $\mathscr{F}$.

By the homogeneity of R, each isomorphism between finite subgraphs can be extended to an automorphism of R, so $Aut(R)$ has as many orbits on n-subsets of $\mathbb{N}$ as the number of non-isomorphic graphs of order n. The number of orbits of $Aut(H)$ on n-subsets is smaller than that, since the orbit of $\{x_1, \ldots, x_n\}$ possibly contains all n-subsets inducing in R a graph that is equivalent, under the relation prescribed by $\mathscr{F}$, to the graph induced by $\{x_1, \ldots, x_n\}$. In any case, there are finitely many orbits on n-subsets of $\mathbb{N}$. Hence R and $H = H(R)$ satisfy the hypothesis of Theorem 2.3.

The next result is a corollary of Theorem 2.3.

Proposition 3.1. *Let $\mathscr{M}$ satisfy the hypothesis of Theorem 2.3 and let T be a subset of $\mathscr{M}^n$ that is invariant under $Aut(\mathscr{M})$. Then there is a formula $\phi(\mathbf{x})$ such that, for any $\mathbf{a} \in \mathscr{M}^n$, $\phi(\mathbf{a})$ is true in $\mathscr{M}$ ($\mathscr{M} \models \phi(\mathbf{a})$) if and only if $\mathbf{a} \in T$.*

Proof. Let $\mathcal{O}_1, \ldots, \mathcal{O}_m$ be the orbits of $Aut(\mathscr{M})$ on n-tuples. Since T is invariant under $Aut(\mathscr{M})$, we can index the orbits so that

$$T = \bigcup_{i=1}^{r} \mathcal{O}_i.$$

Let $\mathbf{x}_1, \ldots \mathbf{x}_r$, $\mathbf{y}_1, \ldots, \mathbf{y}_s$ be representatives of orbits in T and $\mathscr{M} \setminus T$ respectively. By Theorem 2.3, for any i, j $(1 \leq i \leq r, 1 \leq j \leq s)$, there is an n-formula ϕ_{ij} such that

$$\mathscr{M} \models \phi_{ij}(\mathbf{x}_i) \quad \text{and} \quad \mathscr{M} \models \neg\phi_{ij}(\mathbf{y}_j).$$

Set

$$\phi = \bigvee_{i=1}^{r} \bigwedge_{j=1}^{s} \phi_{ij}.$$

Let $\mathbf{a} \in \mathscr{M}^n$. If $\mathbf{a} \in \mathbf{T}$, then $\mathbf{a} \in \mathcal{O}_t$ for some $1 \leq t \leq r$ and $\mathbf{a}$ satisfies the same formulae as $\mathbf{x}_t$. Hence $\mathscr{M} \models \phi(\mathbf{a})$. Conversely, suppose $\mathbf{a} \notin T$. Then $\mathbf{a} \in \mathcal{O}_l$ for some $1 \leq l \leq s$, and for every $1 \leq i \leq r$, $\mathscr{M} \models \neg\phi_{il}(\mathbf{a})$, implying $\mathscr{M} \models \neg\phi(\mathbf{a})$. □

The first-order language of hypergraphs is the same as the first-order language of graphs defined in [1] with adjacency replaced by relations $R_{k_1}, R_{k_2}, \ldots$ (where $k_1, k_2, \ldots$ are the cardinalities of hyperedges), such that $(x_1, \ldots, x_k)$ satisfies R_k if and only if $x_1 \ldots x_k$ is a hyperedge. If a formula ϕ in the language of hypergraphs is true for $H = H(G)$, then the formula ϕ' obtained from ϕ by translating statements like $x_1 \ldots x_k$ *is a hyperedge* into $x_1, \ldots, x_k$ *induces a specified graph* is true for G.

The next four lemmas complete the proof of Theorem 1.1.

Lemma 3.1. *Suppose $\mathscr{F}$ is not invariant under either complementation or switching. Then there is a 2-variable formula $\phi(x, y)$ in the language of hypergraphs, such that for every pair x, y of vertices of R*

$$H \models \phi(x, y) \quad \textit{if and only if} \quad x \sim y \quad \textit{in} \quad R.$$

Proof. Let T be the set of edges of R. Since $Aut(H) = Aut(R)$, T is invariant under $Aut(H)$ and Proposition 3.1 provides the formula. □

If ϕ' is the translation of ϕ, then $R \models (\forall x)(\forall y)(\phi'(x,y) \leftrightarrow x \sim y)$. By Theorem 2.1, this sentence holds in almost all finite graphs. But if it holds in G, then G is determined by $H(G)$: $x \sim y$ if and only if $H(G) \models \phi(x,y)$.

Lemma 3.2. *Suppose $\mathcal{F}$ is invariant under complementation but not under switching. There is a 4-variable formula $\phi(x,y,u,v)$ in the language of hypergraphs, such that for every 4-subset of vertices of R*

$$H \models \phi(x,y,u,v) \quad \textit{if and only if} \quad (x \sim y \wedge u \sim v) \vee (x \not\sim y \wedge u \not\sim v) \quad \textit{in} \quad R.$$

Proof. In Proposition 3.1, let T be the set of 4-tuples (x,y,u,v) for which xy and uv are both edges or both non-edges. Then T is a union of orbits of $Aut(H)$. □

As before, let ϕ' be the translation of ϕ, take two vertices x,y and let u,v be such that $H \models \phi(x,y,u,v)$. By the homogeneity of R, if $x \sim y$ in R, then for every other edge z,w there must be vertices u,v such that $H \models \phi(z,w,u,v)$. Hence, the choice of the first edge determines which of $R, \overline{R}$ will be reconstructed from H when $\mathcal{F}$ satisfies the hypothesis of Lemma 3.2.

Lemma 3.3. *Suppose $\mathcal{F}$ is invariant under switching, but not under complementation. There is a 3-variable formula $\phi(x,y,z)$ in the language of hypergraphs, such that for every 3-subset of R*

$$\begin{aligned} H \models \phi(x,y,z) \quad \textit{if and only if} \quad & (x \sim y \wedge x \not\sim z \wedge y \not\sim z) \vee (x \not\sim y \wedge x \sim z \wedge y \not\sim z) \vee \\ & (x \not\sim y \wedge x \not\sim z \wedge y \sim z) \vee (x \sim y \wedge x \sim z \wedge y \sim z) \\ & \textit{in} \quad R. \end{aligned}$$

Proof. In Proposition 3.1, let T be the set of triples defined by subgraphs of order 3 with odd size. Then T is a union of orbits of $Aut(H)$. □

By Seidel [8], the complete switching class $S(R)$ is determined by knowledge of the triples of vertices that carry an odd number of edges.

Lemma 3.4. *Suppose $\mathcal{F}$ is invariant under switching and complementation. There is a 6-variable formula $\phi(x,y,z,u,v,w)$ in the language of hypergraphs, such that for every 6-tuple of R*

$$H \models \phi(x,y,z,u,v,w)$$

$$\textit{if and only if}$$

$$\big(((x \sim y \wedge x \not\sim z \wedge y \not\sim z) \vee (x \not\sim y \wedge x \sim z \wedge y \not\sim z) \vee (x \not\sim y \wedge x \not\sim z \wedge y \sim z) \vee$$

$$(x \sim y \wedge x \sim z \wedge y \sim z)) \wedge$$

$$((u \sim v \wedge u \not\sim w \wedge v \not\sim w) \vee (u \not\sim v \wedge u \sim w \wedge v \not\sim w) \vee (u \not\sim v \wedge u \not\sim w \wedge v \sim w) \vee$$

$$(u \sim v \wedge u \sim w \wedge v \sim w))\big) \vee$$

$$(((x \not\sim y \wedge x \sim z \wedge y \sim z) \vee (x \sim y \wedge x \not\sim z \wedge y \sim z) \vee (x \sim y \wedge x \sim z \wedge y \not\sim z) \vee$$
$$(x \not\sim y \wedge x \not\sim z \wedge y \not\sim z)) \wedge$$
$$((u \not\sim v \wedge u \sim w \wedge v \sim w) \vee (u \sim v \wedge u \not\sim w \wedge v \sim w) \vee (u \sim v \wedge u \sim w \wedge v \not\sim w) \vee$$
$$(u \not\sim v \wedge u \not\sim w \wedge v \not\sim w))) \quad \textit{in} \quad R.$$

Proof. In Proposition 3.1, let T be the set of 6-tuples (x, y, z, u, v, w) for which the subgraphs induced by x, y, z and u, v, w have both even or both odd order. Then T is a union of orbits of $Aut(H)$. $\square$

4. Examples

Example 1. Set $\mathscr{F} = \{ \; \triangle \; \}$

$H(G)$ determines those edges of G that are contained in a triangle. In the random graph R, every edge has this property. Hence, for almost all graphs G the sentence $(\exists z)(xyz \in H(G))$ holds if and only if $x \sim y$ in G.

Example 2. Set $\mathscr{F} = \{ \; \therefore \; , \; \triangle \; \}$

For this example we have:

Theorem 4.1. *If G, G' have the same $\mathscr{F}$-structure H and $n \geq 10$, then there is a partition H_1, H_2 of H such that H_1 is the set of triples inducing a complete graph in G and G' (or $\overline{G}'$).*

Proof. By Goodman [5], for any $n \geq 6$, the minimum cardinality of H is

$$\left.\begin{array}{ll} 2\binom{n/2}{3} & \text{if } n \text{ is even} \\ \binom{(n-1)/2}{3} + \binom{(n+1)/2}{3} - \frac{n-1}{4} & \text{if } n \equiv 1 \pmod 4 \\ \binom{(n-1)/2}{3} + \binom{(n+1)/2}{3} - \frac{n-3}{4} & \text{if } n \equiv 3 \pmod 4. \end{array}\right\} \qquad (*)$$

Let X be a subset of 6 vertices and let H_X be the subhypergraph of H induced by X. Clearly $H_X = \mathscr{F}(G_X) = \mathscr{F}(G'_X)$. Suppose the set of triangles of G_X is neither the set of triangles of G'_X nor that of $\overline{G}'_X$. We can assume there exist two triples $h_1 = xyz$, $h_2 = uvw$ of H_X such that h_1 induces a triangle in both G_X and G'_X, and h_2 induces a triangle in G_X and a null-graph in G'_X, so $| h_1 \cap h_2 | \leq 1$.

Claim 1. $| h_1 \cap h_2 | = 1$.

Proof. Suppose $| h_1 \cap h_2 | = 0$.

Case 1. $H_X = \{h_1, h_2\}$. At least one of xu, xv, yu, yv, zu, zv must be an edge in G'_X. Without loss, assume $x \sim u$. Then G'_X looks like the graph in Figure 1 (dotted lines are the forced non-edges). So, H_X contains at least one more hyperedge.

Case 2. (a) $xyu \in H_X$ (b) $xuv \in H_X$

Suppose (a) occurs. Then xyu is a triangle in both G_X and G'_X, which forces

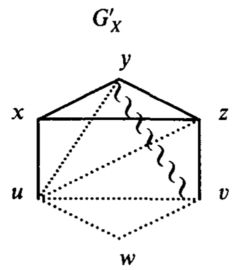

Figure 1

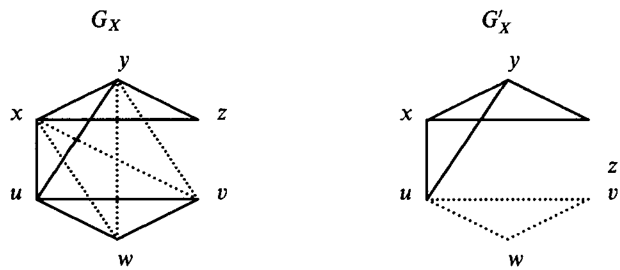

Figure 2

$x \not\sim v, x \not\sim w, y \not\sim v, y \not\sim w$ in G_X. As shown in Figure 2, x and y must be joined to v or w in G'_X. By symmetry, we can assume $y \sim w$, which forces $z \not\sim w, x \not\sim w$. So, G'_X and therefore G_X, is determined except for the pair uz.

Figure 3 shows that no choice for uz can produce two graphs with the same $\mathscr{F}$-structure.

Case (b) is similarly impossible. □

Now suppose Theorem 4.1 is false. Let H_1, H_2 be the partition of H into triangles and null-graphs of G, and let H'_1, H'_2 be the analogous partition for G'.

Claim 2. *Let* $H = \{h_1, h_2, \ldots, h_m\}$. *For any distinct* $1 \leq i, j \leq m$, $\quad |\, h_i \cap h_j \,| \leq 1$.

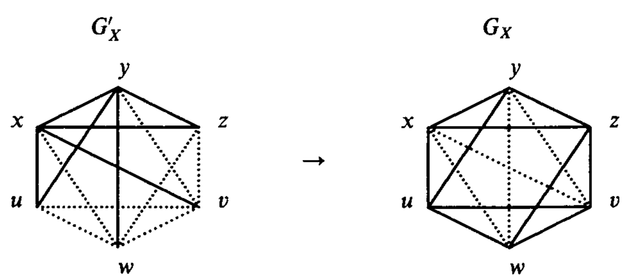

Figure 3

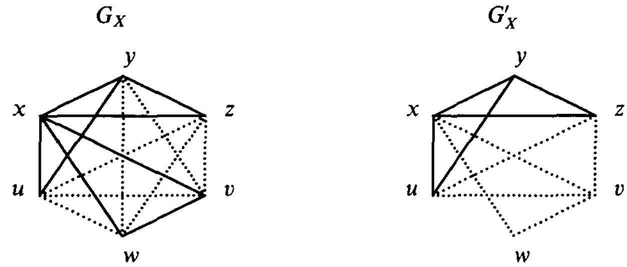

Figure 4

Proof. Suppose for some i, j that $| h_i \cap h_j | = 2$, say $h_i = xyz, h_j = xyw$. Then h_i, h_j are both in the same partition. We can assume $h_i, h_j \in H_1 \cap H_2$ (otherwise take the complement). Let h_k be a member of $H_1 \setminus H'_1$. By Claim 1, $| h_k \cap h_i | = | h_k \cap h_j | = 1$. It follows that $xyzw$ induces a $K_4 - e$ in both G and G'. Therefore, $h_k = xuv$ or yuv. If $h_k = xuv$, then $vwz \in H_2 \cap H'_2$. The subgraph of G induced by $X = \{x, y, z, u, v, w\}$ is pictured in Figure 4, and any attempt to define the edges of G'_X fails to produce the same $\mathcal{F}$-structure as G_X. □

Claim 2 implies $| H | \leq \binom{n}{2}/3$. Since there is no number satisfying

$$\frac{\binom{n}{2}}{3} \geq | H | \geq (*),$$

the proof of Theorem 4.1 is complete. □

For the collection $\mathcal{F}$ of Example 2, every graph with the property that each pair of vertices lies either in a triangle or in a 'null-triangle' is determined up to complementation by its $\mathcal{F}$-structure.

Consider the line graph of $K_{3,3}$. Then the $\mathcal{F}$-structure is the affine plane or Steiner system $AG(2, 3)$. For any choice of two parallel classes, there is a unique graph G such that the lines in the chosen classes are the triangles in G and the other lines are null-triangles. So there are $\binom{4}{2}/2 = 3$ ways of partitioning the hyperedges so that the conclusion of Theorem 4.1 holds.

5. Acknowledgement

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References

[1] Blass, A. and Harary, F. (1979) Properties of Almost all Graphs and Complexes. *J. Graph Theory* **3** 225–240.

[2] Cameron, P. J. (1984) Aspects of the Random Graph. In: Bollobás, B. (ed.) *Graph Theory and Combinatorics*, Acad. Press 65–79.

[3] Cameron, P. J. (1990) Oligomorphic Permutation Groups. *London Math. Soc. Lecture Note Series* **152**, Cambridge University Press.
[4] Fagin, R. (1976) Probabilities on Finite Models. *J. Symbolic Logic* **41** (1) 50–58.
[5] Goodman, A. W. (1959) On Sets of Acquaintances and Strangers at Any Party. *Amer. Math. Monthly* **66** 778–783.
[6] Reed, B. (1987) A Semi-Strong Perfect Graph Theorem. *J. Combinatorial Theory B* **43** 223–240.
[7] Rado, R. (1964) Universal Graphs and Universal Functions. *Acta Arith.* **9** 331–340.
[8] Seidel, J. J. (1973) A Survey of Two-Graphs. *Accademia Nazionale dei Lincei Atti dei Convegni Lincei 17*, Colloquio Internazionale Sulle Teorie Combinatorie, Roma.
[9] Thomas, S. (1991) Reducts of the Random Graph. *J. Symbolic Logic* **56** 176–181.